

Kingdom Fungi: A Comprehensive Review for NEET-UG



Morphology, Reproduction, and Classification (NCERT Based)

Target: NEET-UG | Focus: Statement-Based Mastery

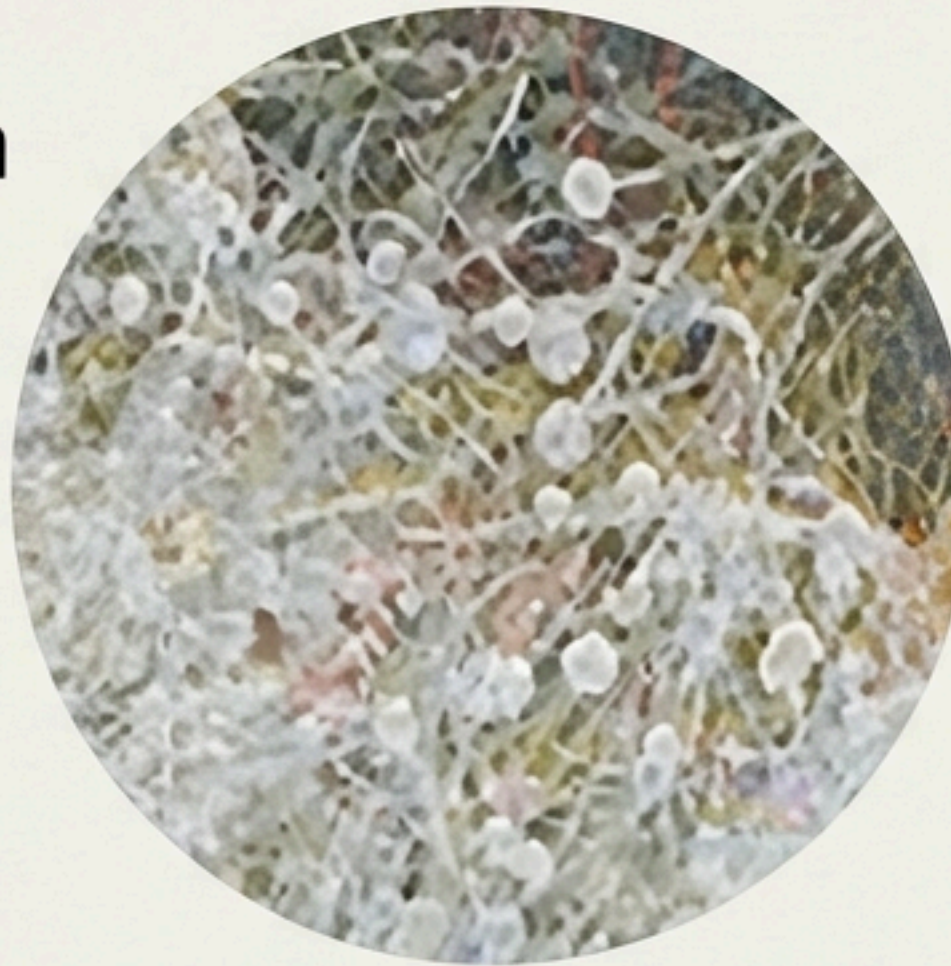
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General Characteristics & Habitat

Heterotrophic: A unique kingdom of heterotrophic organisms.

Cosmopolitan: Found in air, water, soil, animals, and plants.

Preference: Grow in warm and humid places (e.g., spoilage prevented by refrigeration).



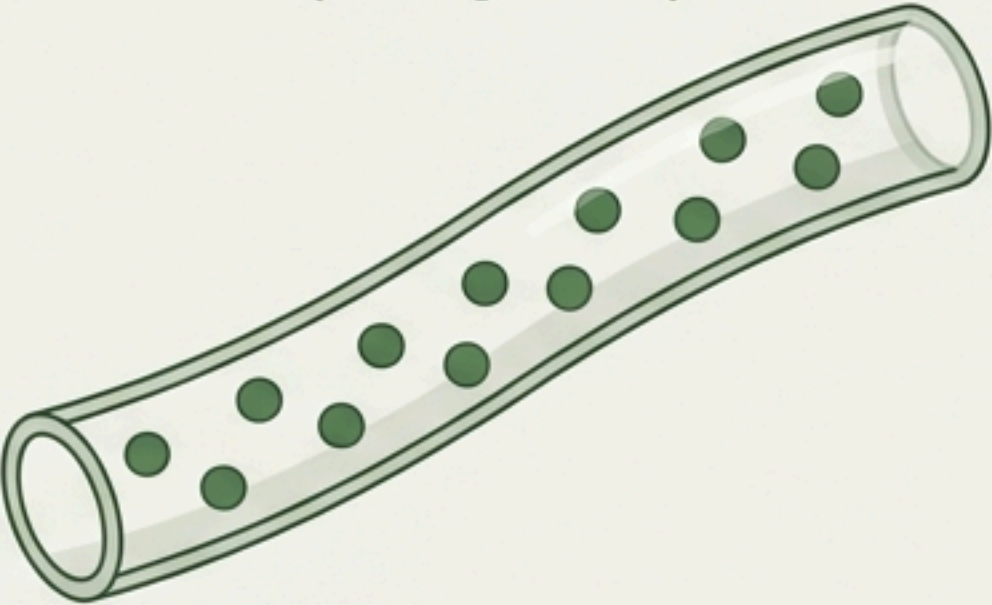
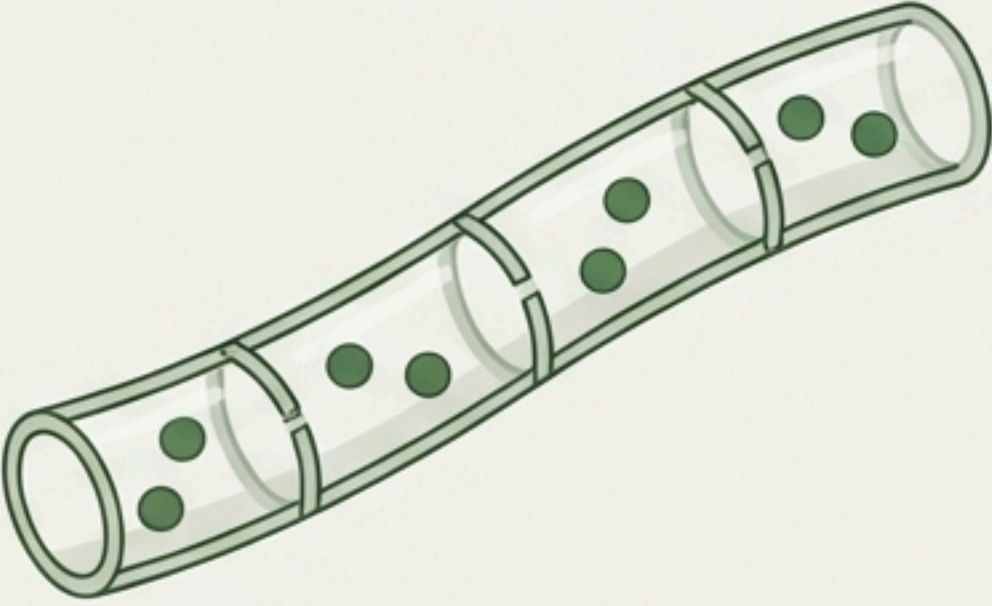
NEET ALERT: KEY STATEMENTS

- White spots seen on mustard leaves are due to a parasitic fungus.
- Cell walls of fungi are composed of **chitin and polysaccharides**.
- With the exception of yeasts (unicellular), fungi are filamentous.

Structure: Hyphae & Mycelium

Hyphae: Long, slender thread-like structures.

Mycelium: The network of hyphae.

Hyphae Types	
A Coenocytic Hyphae (Aseptate) 	B Septate Hyphae 

NEET ALERT: KEY STATEMENTS

- Some hyphae are continuous tubes filled with multinucleated cytoplasm – these are called **coenocytic hyphae**.

Modes of Nutrition

Saprophytes

Absorb soluble organic matter from dead substrates.



Parasites

Depend on living plants and animals.



Symbionts

- **Lichens:** Association with algae.
- **Mycorrhiza:** Association with roots of higher plants.



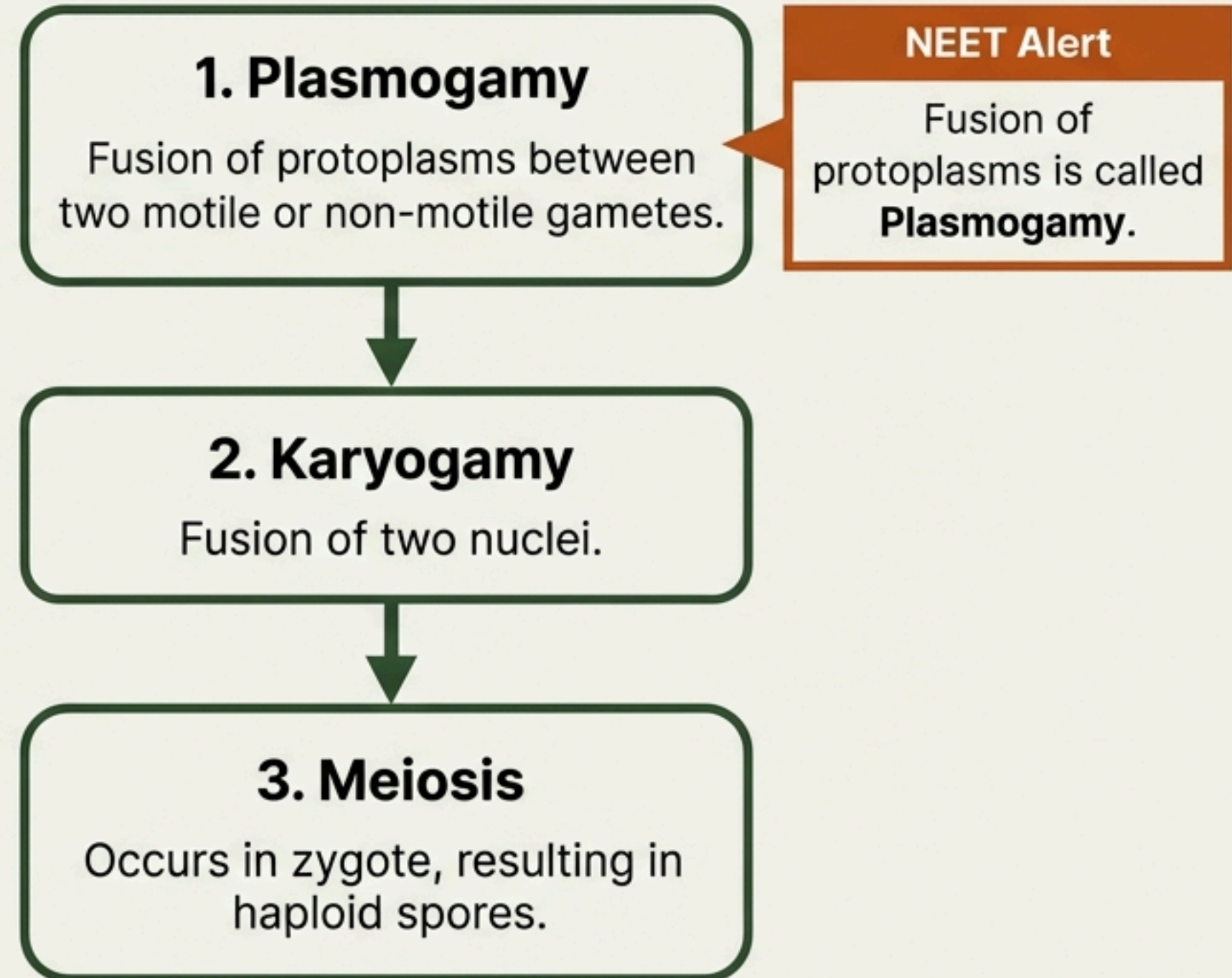
STATEMENT FOCUS: Most fungi are heterotrophic and absorb soluble organic matter from dead substrates (saprophytes).

Reproduction: The Sexual Cycle

Vegetative Means:

Fragmentation, fission, budding.

Asexual Spores: Conidia,
Sporangiospores, Zoospores.



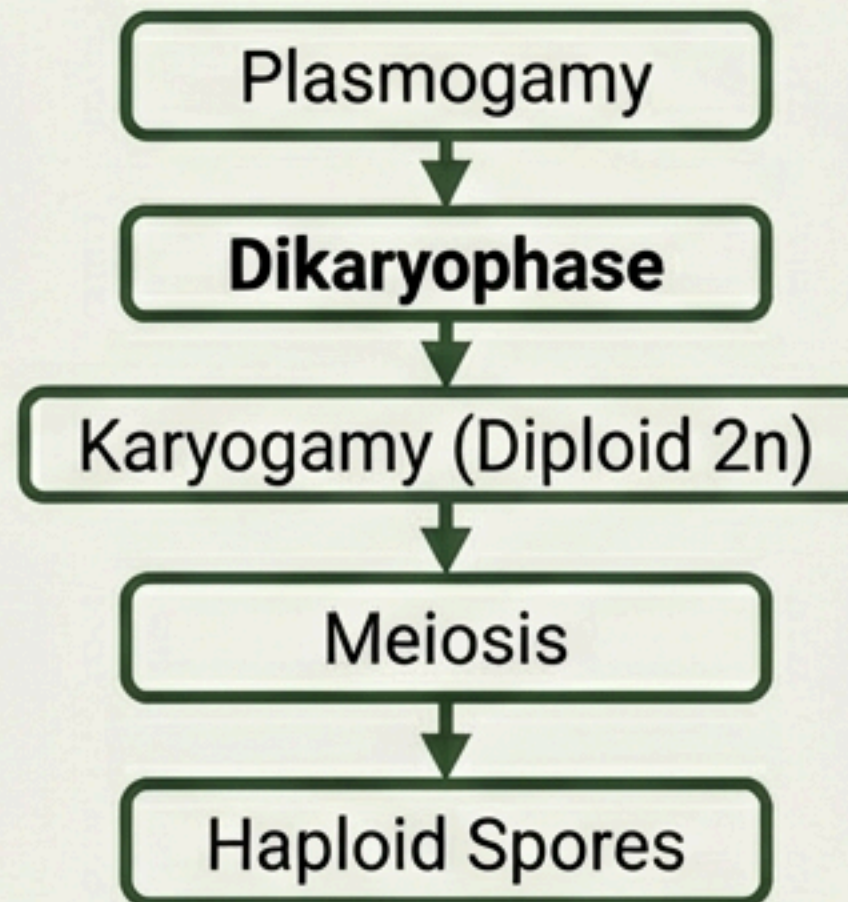
The 'Dikaryon' Concept ($n + n$)

Occurs in **Ascomycetes** and **Basidiomycetes**.

$$n + n$$

Two nuclei per cell (Dikaryon stage).

Fusion is not immediate.

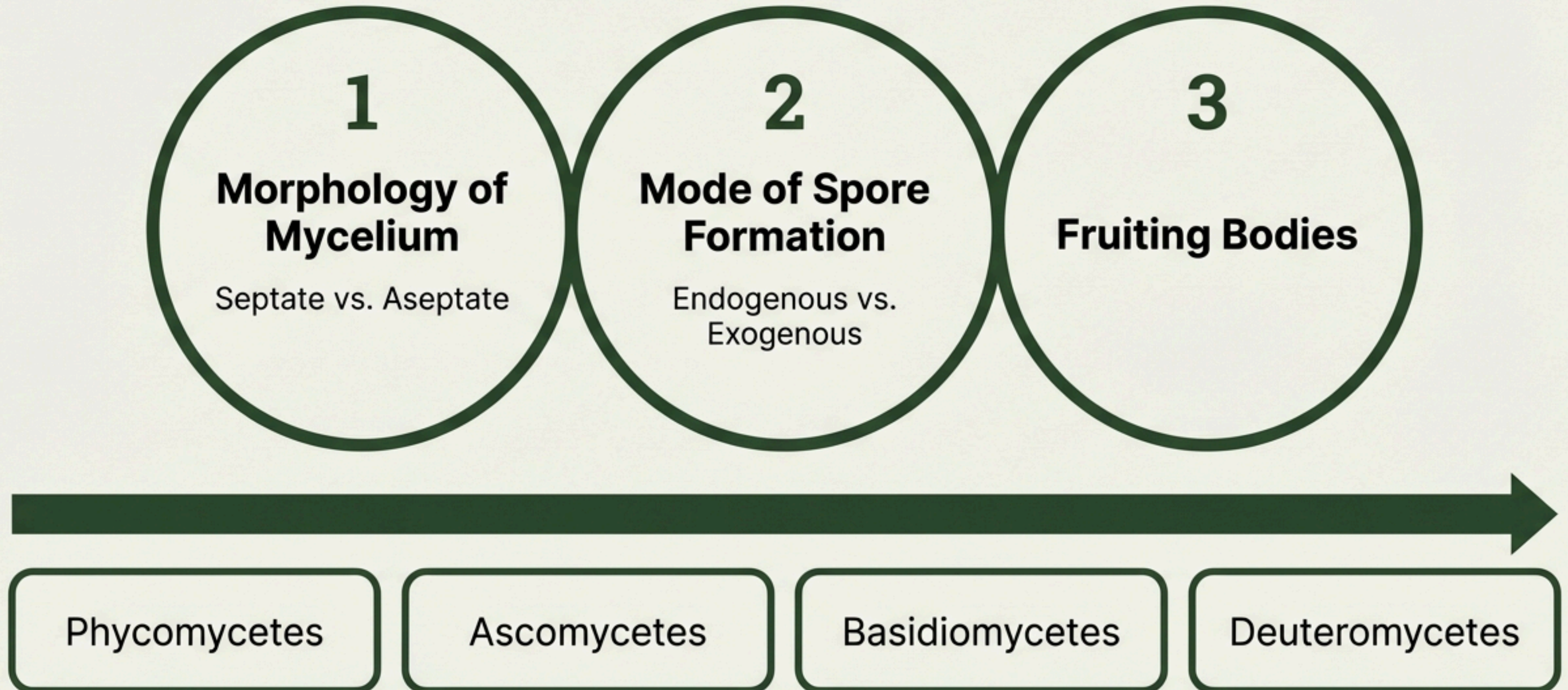


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- In ascomycetes and basidiomycetes, an intervening dikaryotic stage occurs.
- Later, the parental nuclei fuse and the cells become diploid.

Basis of Classification

The Kingdom is divided into classes based on three criteria:



Class 1: Phycomycetes (Lower Fungi)

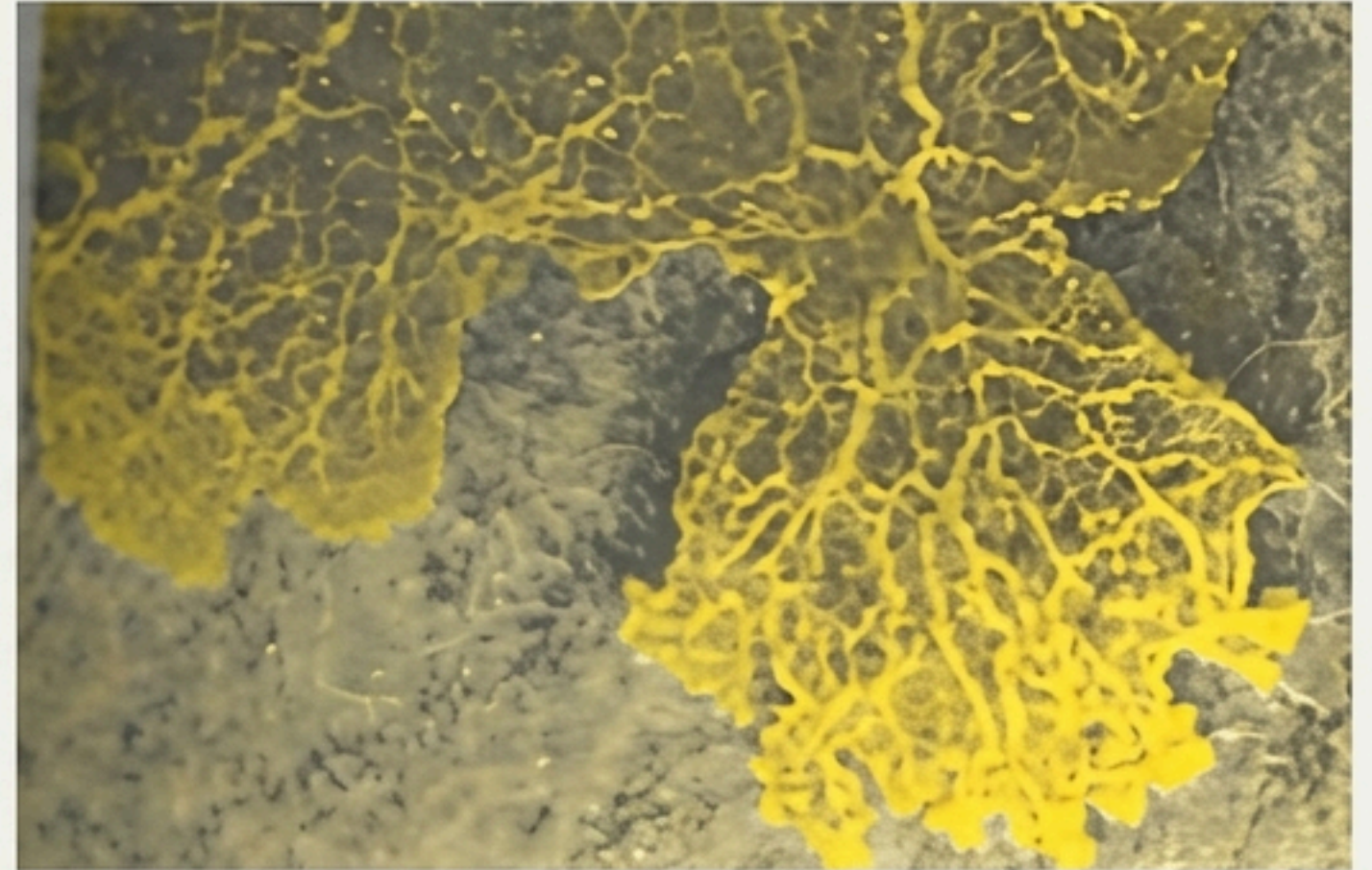
Habitat: Aquatic, decaying wood (moist/damp), obligate parasites on plants.

Mycelium: Aseptate and Coenocytic.

Asexual Repro: Zoospores (motile) or Aplanospores (non-motile). Produced **Endogenously** in sporangium.

Sexual Repro: Zygosporangium formed by fusion (Isogamous/Anisogamous/Oogamous).

Examples: *Mucor*, *Rhizopus* (Bread mould), *Albugo* (Parasitic on mustard).



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CRITICAL: The mycelium is aseptate and coenocytic.

Class 2: Ascomycetes (Sac-Fungi)

Lifestyle: Saprophytic, decomposers, parasitic, or coprophilous (growing on dung).

Mycelium: Branched and **Septate**.

Asexual Repro: Conidia produced **Exogenously** on conidiophores.

Sexual Repro: Ascospores produced **Endogenously** in Asci (Ascocarps).

Examples: *Penicillium*, *Yeast*, *Aspergillus*, *Claviceps*, *Neurospora*.



Aspergillus
(Fan-like Head)



Yeast
(Cellular Structure)

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- *Neurospora* is used extensively in biochemical/genetic work.
- Morels and truffles are edible delicacies.

Class 3: Basidiomycetes (Mushrooms)

Habitat: Soil, logs, parasites (**Rusts and Smuts**).

Mycelium: Branched and Septate.

Key Features:

- Asexual spores generally **absent**.
- **No sex organs**. Plasmogamy by fusion of somatic cells.
- **Basidiospores** produced **Exogenously** on Basidium.

Examples: *Agaricus* (Mushroom), *Ustilago* (Smut), *Puccinia* (Rust fungus).



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- Sex organs are absent.
- Plasmogamy is brought about by fusion of two vegetative or somatic cells.

Class 4: Deuteromycetes (Imperfect Fungi)

Why Imperfect?

Only asexual or vegetative phases are known. Once sexual stages are discovered, they are moved to Ascomycetes or Basidiomycetes.

Characteristics:

- Reproduction: Only by asexual spores (**Conidia**).
- Mycelium: Septate and branched.

Role:

Decomposers of litter and help in **mineral cycling**.

Examples:

Alternaria, Colletotrichum, Trichoderma.

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Large number of them are decomposers of litter and help in mineral cycling.

Comparative Review: The 4 Classes

Class	Mycelium	Asexual Spore	Sexual Spore	Examples
Phycomycetes	Aseptate / Coenocytic	Zoo/Aplano (Endogenous)	Zygospore	<i>Mucor, Albugo</i>
Ascomycetes	Septate / Branched	Conidia (Exogenous)	Ascospore (Endogenous)	<i>Neurospora, Yeast</i>
Basidiomycetes	Septate / Branched	Absent	Basidiospore (Exogenous)	<i>Agaricus, Puccinia</i>
Deuteromycetes	Septate / Branched	Conidia	Absent / Unknown	<i>Trichoderma</i>

NEET Challenge: Hard Statement Questions (1-5)

Q1: Which class of fungi is characterized by aseptate and coenocytic mycelium?
(A) Basidiomycetes (B) Ascomycetes (C) Phycomycetes (D) Deuteromycetes

Q2: Identify the **incorrect** statement:

- (A) Yeast is unicellular.
- (B) Fungal cell walls are made of chitin.
- (C) *Albugo* is a parasitic fungus on mustard.
- (D) Coenocytic hyphae have septae.

Q3: In which class are sex organs absent, but plasmogamy occurs by fusion of somatic cells?

- (A) Basidiomycetes (B) Ascomycetes (C) Phycomycetes
- (D) Deuteromycetes (D) Deuteromycetes

Q4: 'Coprophilous' fungi grow on:

- (A) Wood (B) Dung (C) Living Plants (D) Rocks

Q5: Where are Ascospores and Basidiospores produced, respectively?

- (A) Endogenously, Exogenously (B) Exogenously, Endogenously
- (C) Both Endogenously (D) Both Exogenously

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Read the questions carefully, especially when asked to identify the **incorrect** statement. Pay close attention to terms like 'aseptate', 'coenocytic', and spore formation types.

NEET Challenge: Assertion & Matching (6-10)

Q6 (Assertion/Reason):

Assertion: Deuteromycetes are called imperfect fungi.

Reason: Only asexual or vegetative phases are known.

Q7 (Matching): Match the fungus to its role/feature:

- | | |
|--------------------------|--------------------------|
| (a) <i>Puccinia</i> | (i) Biochemical genetics |
| (b) <i>Neurospora</i> | (ii) Rust fungus |
| (c) <i>Saccharomyces</i> | (iii) Bread making |

Q8: Which stage is represented by $n + n$?

- (A) Monokaryon (B) Dikaryon (C) Zygote (D) Coenocytic

Q9: Conidia in Ascomycetes are produced _____, whereas spores in Phycomycetes are produced _____.

Q10: Which grouping is exclusively distinct by having *no* known sexual stage?

- (A) Phycomycetes (B) Deuteromycetes (C) Basidiomycetes (D) Ascomycetes

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Pay close attention to terms like '**Assertion**', '**Reason**', and '**Exogenously/Endogenously**'. Matching questions require precise knowledge of specific examples and their roles. Ensure you understand the definitions of $n + n$ (dikaryon) and the characteristics of each fungal class.

Answer Key & Final Thoughts

Answer Key

- 1: (C) Phycomycetes
- 2: (D) Incorrect because Coenocytic means aseptate (no septae).
- 3: Basidiomycetes
- 4: (B) Dung
- 5: (A) Endo (Asco) / Exo (Basidio)
- 6: Both True & Correct Explanation
- 7: a-ii, b-i, c-iii
- 8: (B) Dikaryon
- 9: Exogenously / Endogenously
- 10: (B) Deuteromycetes

Closing Thoughts

Summary: Mastering Kingdom Fungi requires tracking the exceptions: The unicellular Yeast, the aseptate Phycomycetes, and the sexual-stage-lacking Deuteromycetes.

Remember the location of spores (Endo vs Exo) for high-yield success.

